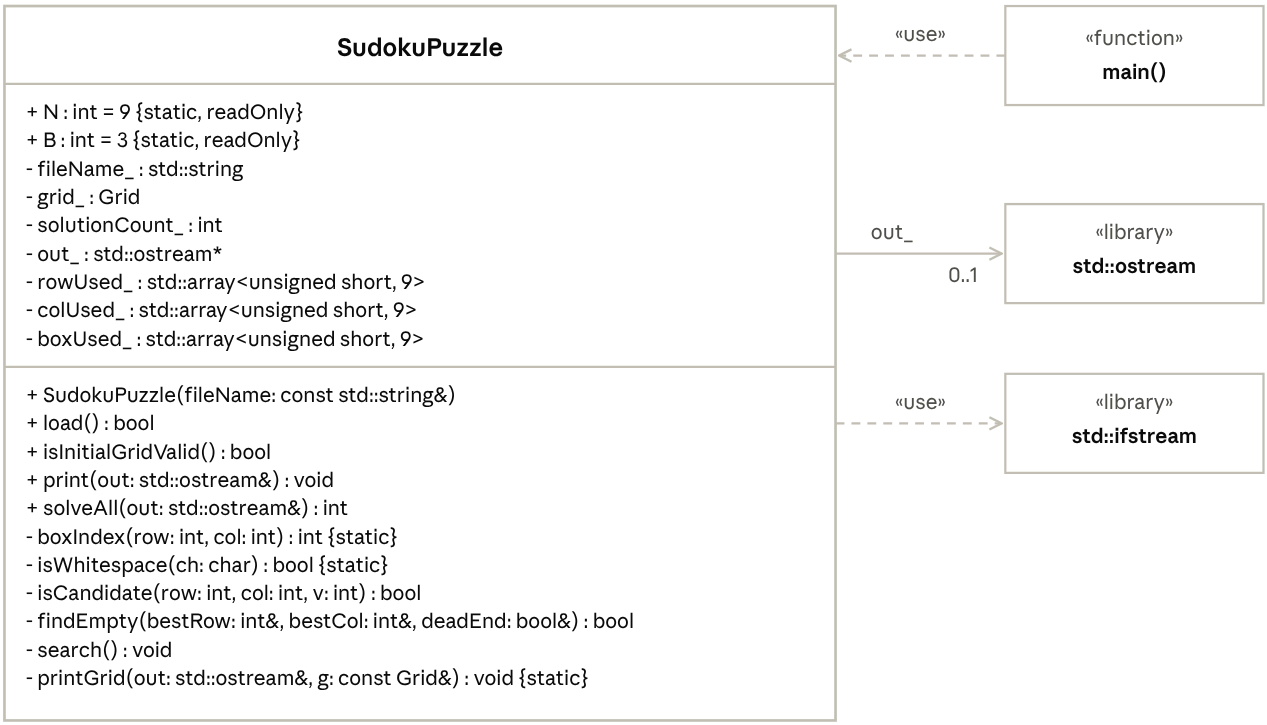
UML Class Diagram: Sudoku Solver

Assignment 6

Classes

The program has one user-defined class, SudokuPuzzle. main() is a free function, not a class, so it appears only as the code that uses SudokuPuzzle.

One class, SudokuPuzzle, does the loading, solving and printing



+ public, - private. Grid = std::array<std::array<int, 9>, 9>. Solid arrow: association. Dashed arrow: dependency.

class diagram · 1 class, 3 relationships

SudokuPuzzle

Visibility: + public, - private. Members marked {static} belong to the class, not to one object.

Attributes

Visibility	Name	Type	Description
+	N	int = 9 {static, readOnly}	Size of the grid
+	B	int = 3 {static, readOnly}	Size of each block
-	fileName_	std::string	Name of the puzzle file
-	grid_	Grid	The 9x9 puzzle grid, 0 means empty. Grid is a private alias for std::array<std::array<int, 9>, 9>
-	solutionCount_	int	Number of solutions found so far
-	out_	std::ostream*	Stream the solutions are written to (not owned)
-	rowUsed_	std::array<unsigned short, 9>	Bitmask of digits already used in each row
-	colUsed_	std::array<unsigned short, 9>	Bitmask of digits already used in each column
-	boxUsed_	std::array<unsigned short, 9>	Bitmask of digits already used in each 3x3 box

Methods

Visibility	Name	Parameter list	Return type	Description
+	SudokuPuzzle (constructor)	fileName: const std::string&	none	Stores the puzzle file name
+	load	none	bool	Reads the 81 cells from the file; false if the file is unreadable or malformed
+	isInitialGridValid	none	bool	Checks the clues for conflicts and fills the bitmasks

Visibility	Name	Parameter list	Return type	Description
+	print	out: std::ostream&	void	Writes the current grid to a stream
+	solveAll	out: std::ostream&	int	Finds and writes every solution; returns how many
-	boxIndex {static}	row: int, col: int	int	Converts a cell position to its 3x3 box number (0-8)
-	isWhitespace {static}	ch: char	bool	True for space, newline, carriage return and tab
-	isCandidate	row: int, col: int, v: int	bool	True if digit v can legally go in the cell
-	findEmpty	bestRow: int&, bestCol: int&, deadEnd: bool&	bool	Picks the empty cell with the fewest candidates; false if the grid is full
-	search	none	void	Recursive depth-first search with backtracking
-	printGrid {static}	out: std::ostream&, g: const Grid&	void	Writes a grid in the output format

Relationships

- main() to SudokuPuzzle: dependency. main() creates one SudokuPuzzle for each puzzle file and calls load(), isInitialGridValid() and solveAll().
- SudokuPuzzle to std::ostream: association through out_ (0..1, not owned). solveAll() stores the stream it is given, and search() writes each solution through it.
- SudokuPuzzle to std::ifstream: dependency. load() opens a local std::ifstream to read the puzzle file.

std::string and std::array appear as attribute types, so they are not drawn as separate classes.